



Sistem Informasi Geografis



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SIG – Remote Sensing Indices

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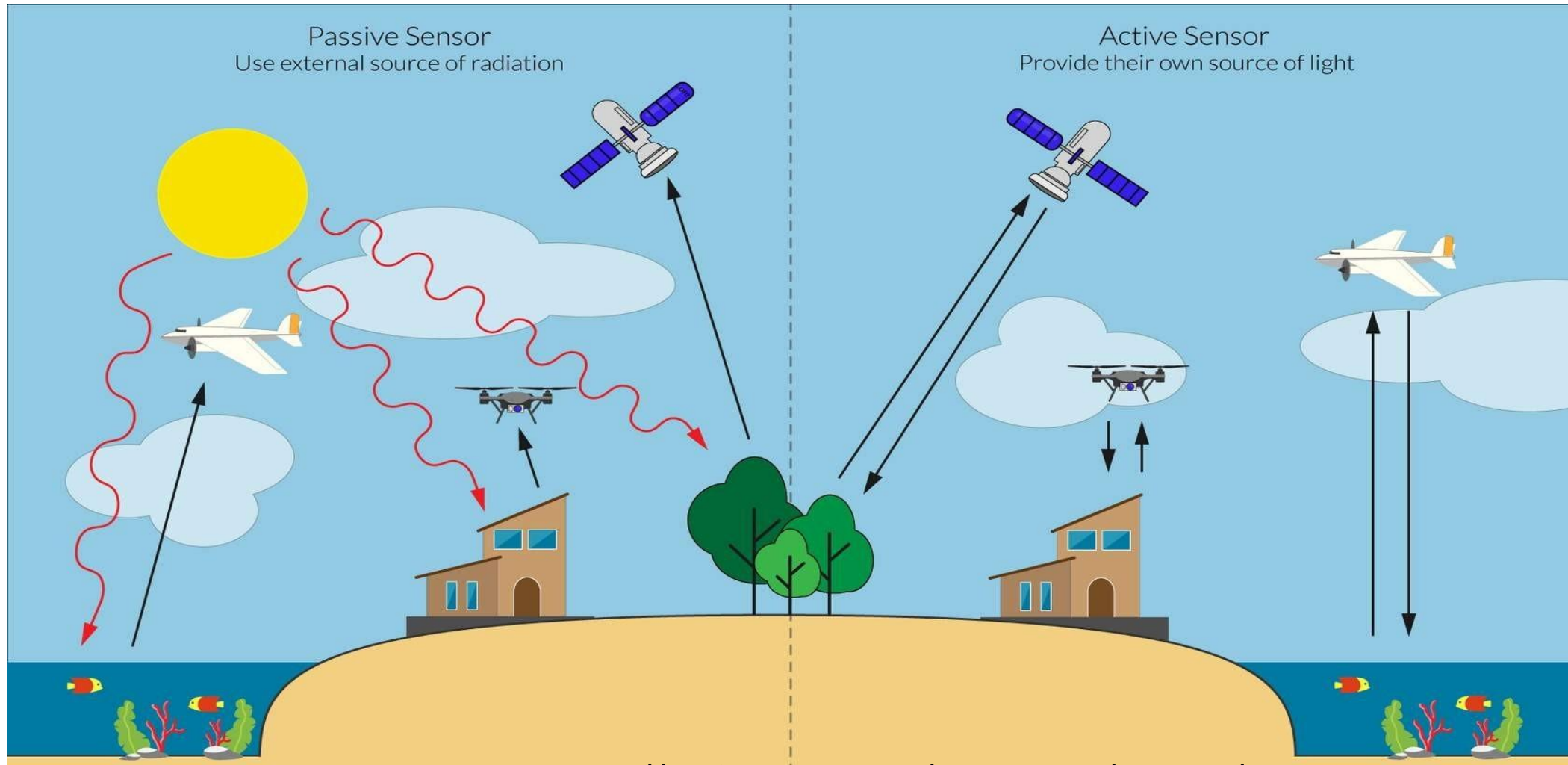


   @rojulman

Geographical Information System



Jenis Sensor Penginderaan Jauh

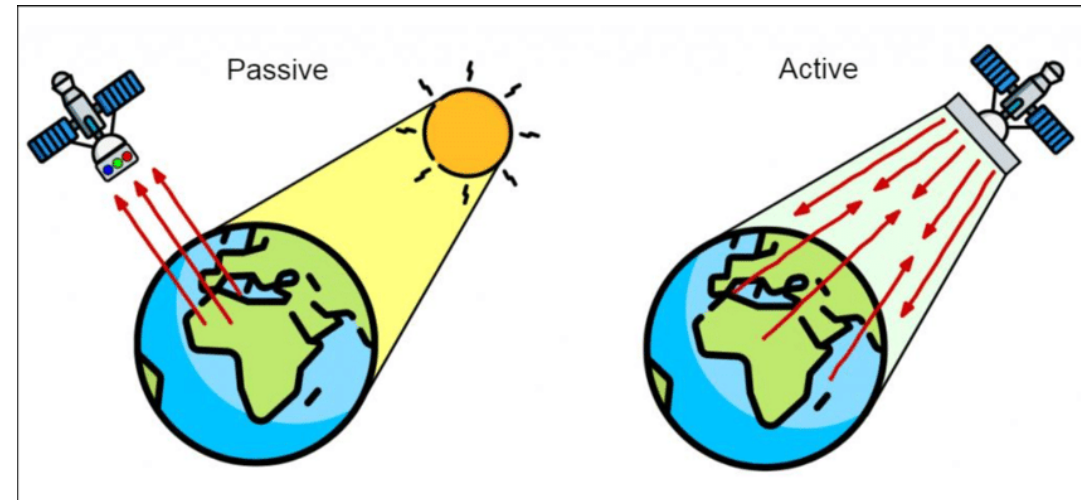


<https://reefresilience.org/wp-content/uploads/passive-and-active-sensors.jpg>

Jenis Sensor Penginderaan Jauh

- **Sensor Pasif**

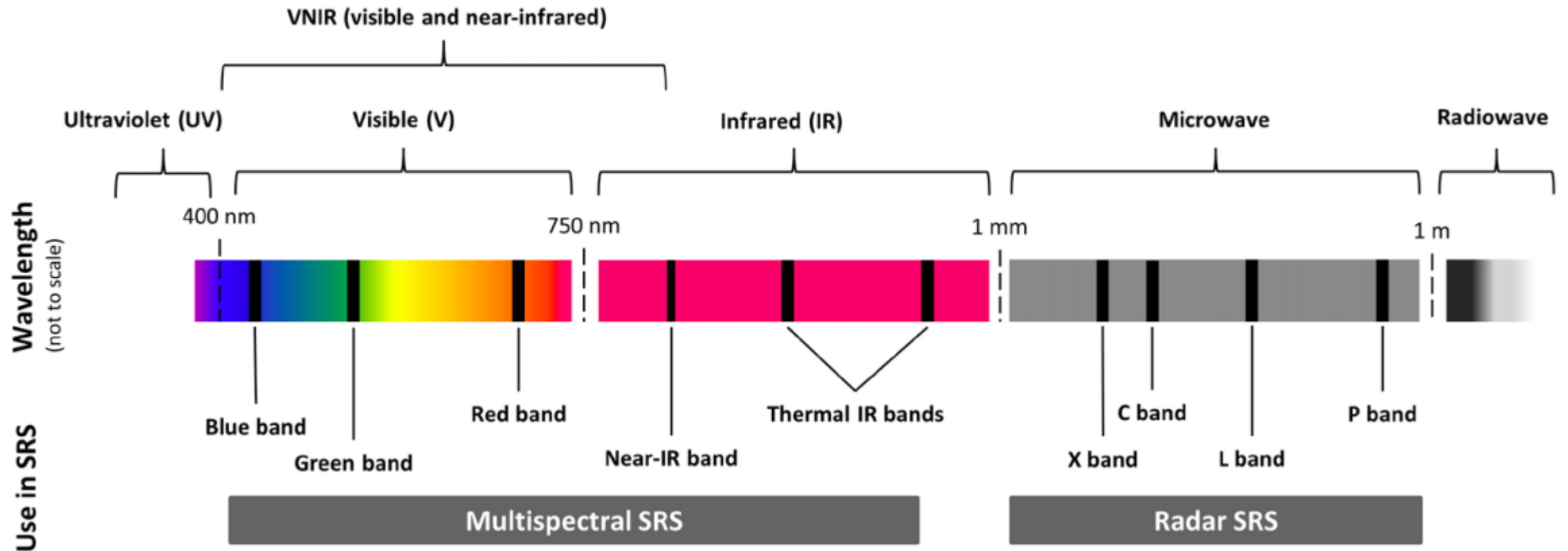
- Sumber energinya berasal dari eksternal, utamanya dari matahari
- Perekaman hanya dapat dilakukan siang hari
- Menampilkan kondisi permukaan bumi yang sesuai kenyataannya dengan menangkap gelombang pendek



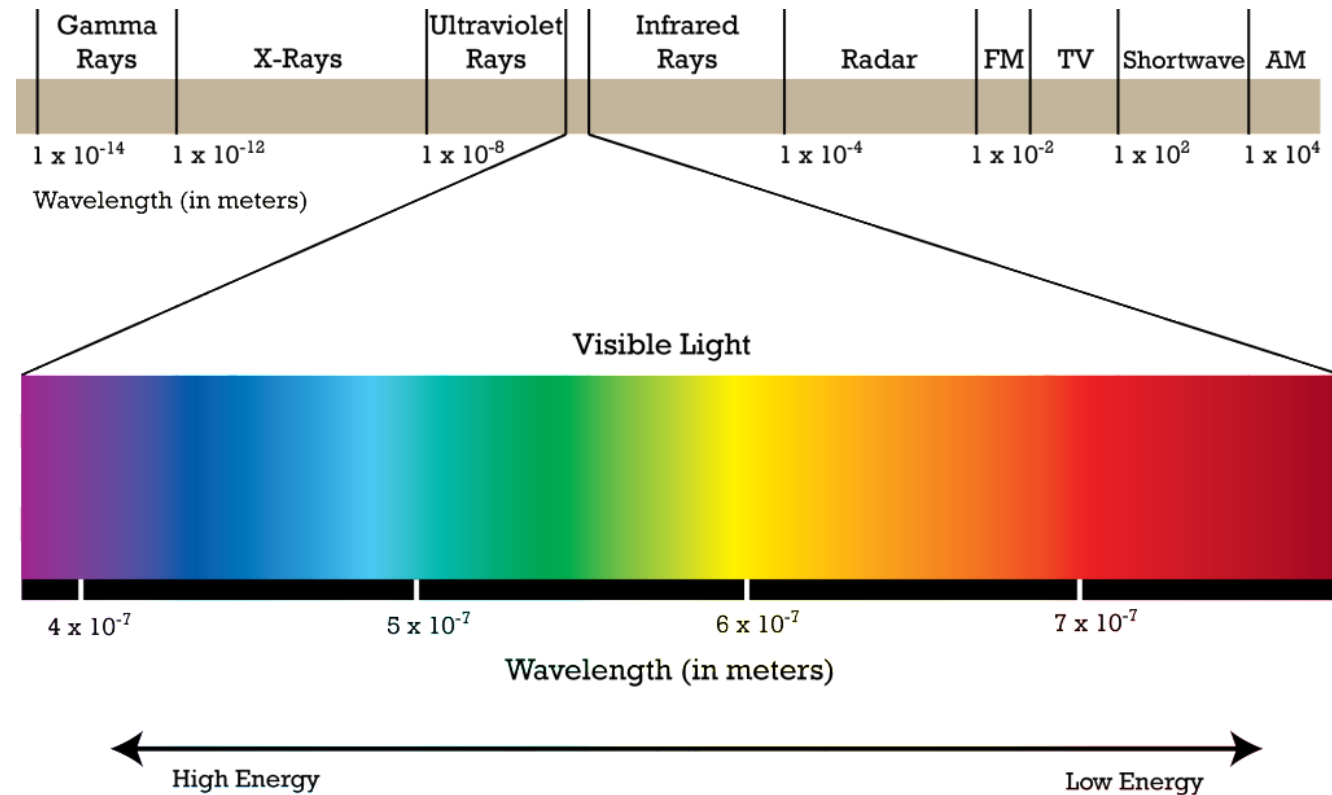
- **Sensor Aktif**

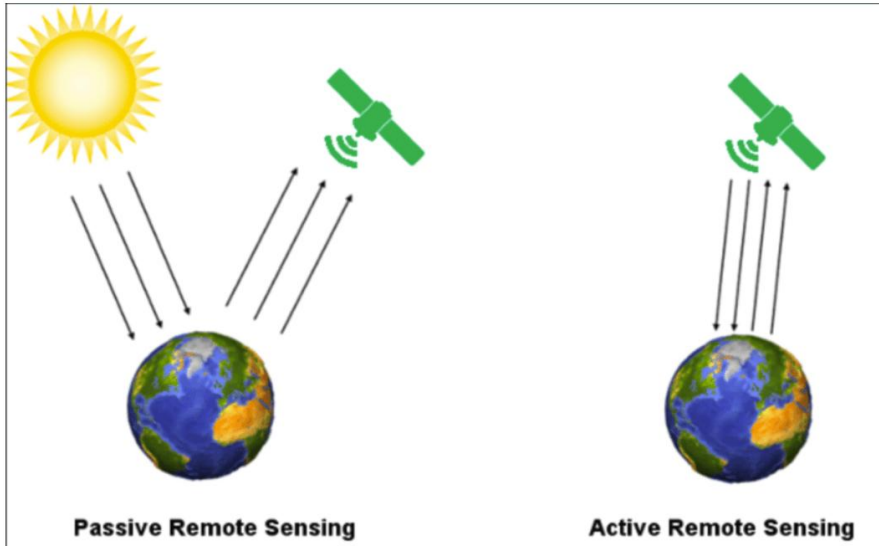
- Memiliki sumber energi sendiri
- Perekaman dapat dilakukan kapan saja, termasuk malam hari
- Memanfaatkan gelombang panjang yang dapat menembus awan

Band Satelit - Multispectral vs Radar



- Sensor Remote Sensing dapat mengumpulkan data semua bagian dari spektrum gelombang Electro Magnetic
- Sensor multispectral memiliki batasan sensitivitas spektral (resolusi spektral)
- Rentang panjang gelombang yang direkam sensor disebut “PITA” / “BAND” / “CHANNEL” yang bervariasi



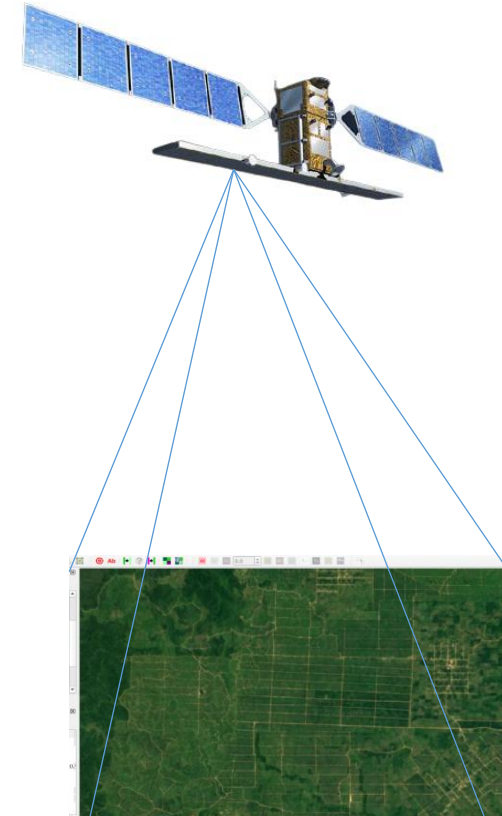


❖ **C-SAR Sentinel 1:** pencitraan radar yang mengorbit, semua cuaca, siang dan malam untuk layanan darat dan laut.

Product:
Sentinel-1
GRD Level-1

Parameter	Description
Polarization	Dual (HH+HV, VV+VH)
Swath	250
Ground range resolution	5 m
Incident Angle	31° - 46°
Azimuth resolution	20 m
Azimuth & Range looks	Single
Maximum noise-equivalent sigma zero	-22 dB
Radiometric stability	0.5 dB (3 σ)
Radiometric accuracy	1 dB (3 σ)
Phase error	5°

2



❖ Sentinel 2 – Multispectral:

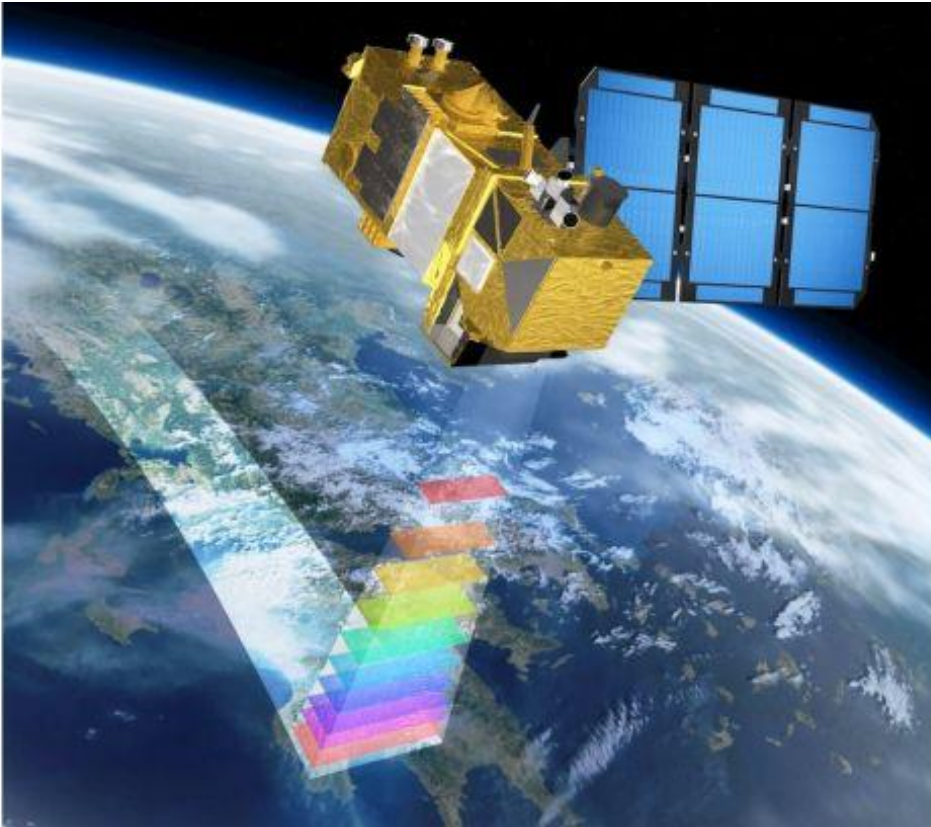
- Multi Spectral Imaginary (MSI) covering 13 spectral bands (443–2190 nm)
- A swath width of 290 km and a spatial resolution of
 - 10 m (four visible and near-infrared bands),
 - 20 m (six red edge and shortwave infrared bands)
 - 60 m (three atmospheric correction bands).

Sentinel-2 Bands	Central Wavelength (μm)	Resolution (m)
Band 1 - Coastal aerosol	0.443	60
Band 2 - Blue	0.490	10
Band 3 - Green	0.560	10
Band 4 - Red	0.665	10
Band 5 - Vegetation Red Edge	0.705	20
Band 6 - Vegetation Red Edge	0.740	20
Band 7 - Vegetation Red Edge	0.783	20
Band 8 - NIR	0.842	10
Band 8A - Vegetation Red Edge	0.865	20
Band 9 - Water vapour	0.945	60
Band 10 - SWIR - Cirrus	1.375	60
Band 11 - SWIR	1.610	20
Band 12 - SWIR	2.190	20

13

(Sumber: Sentinel 2 Handbook)

Band – Sentinel 2A



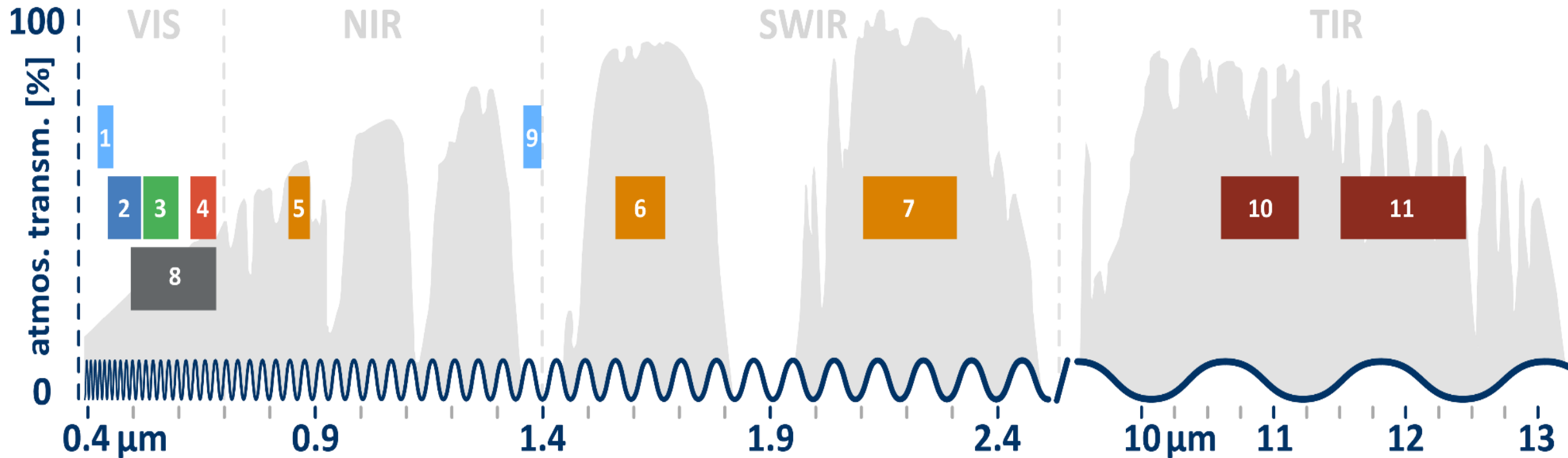
ESA – European Space Agency

13

	Band	Spectral /wavelength range (nm)	Objective	Spatial range (m)
B1	Coastal aerosol	433-453	Aerosol correction	60
B2	Blue	458-523	Aerosol correction, land measurement	10
B3	Green	543-578	Land measurement	10
B4	Red	650-680	Land measurement	10
B5	Red edge1 (RE1)	698-713	Land measurement	20
B6	Red edge2 (RE2)	733-748	Land measurement	20
B7	Red edge3 (RE3)	773-793	Land measurement	20
B8	Near infra red	785-900	Water vapour correction, Land measurement	10
B8a	Near infrared narrow	855-875	Water vapour correction, Land measurement	20
B9	Water vapour	935-955	Water vapour correction	60
B10	Shortwave infrared	1360-1390	Cirrus detection	60
B11	Shortwave infrared 1	1565-1655	Land measurement	20
B12	Shortwave infrared 2	2100-2280	Aerosol correction, land measurement	20

Band Satelit - Landsat

- Satelit Landsat mengumpulkan data dalam 11 “channels” di seluruh spektrum EM

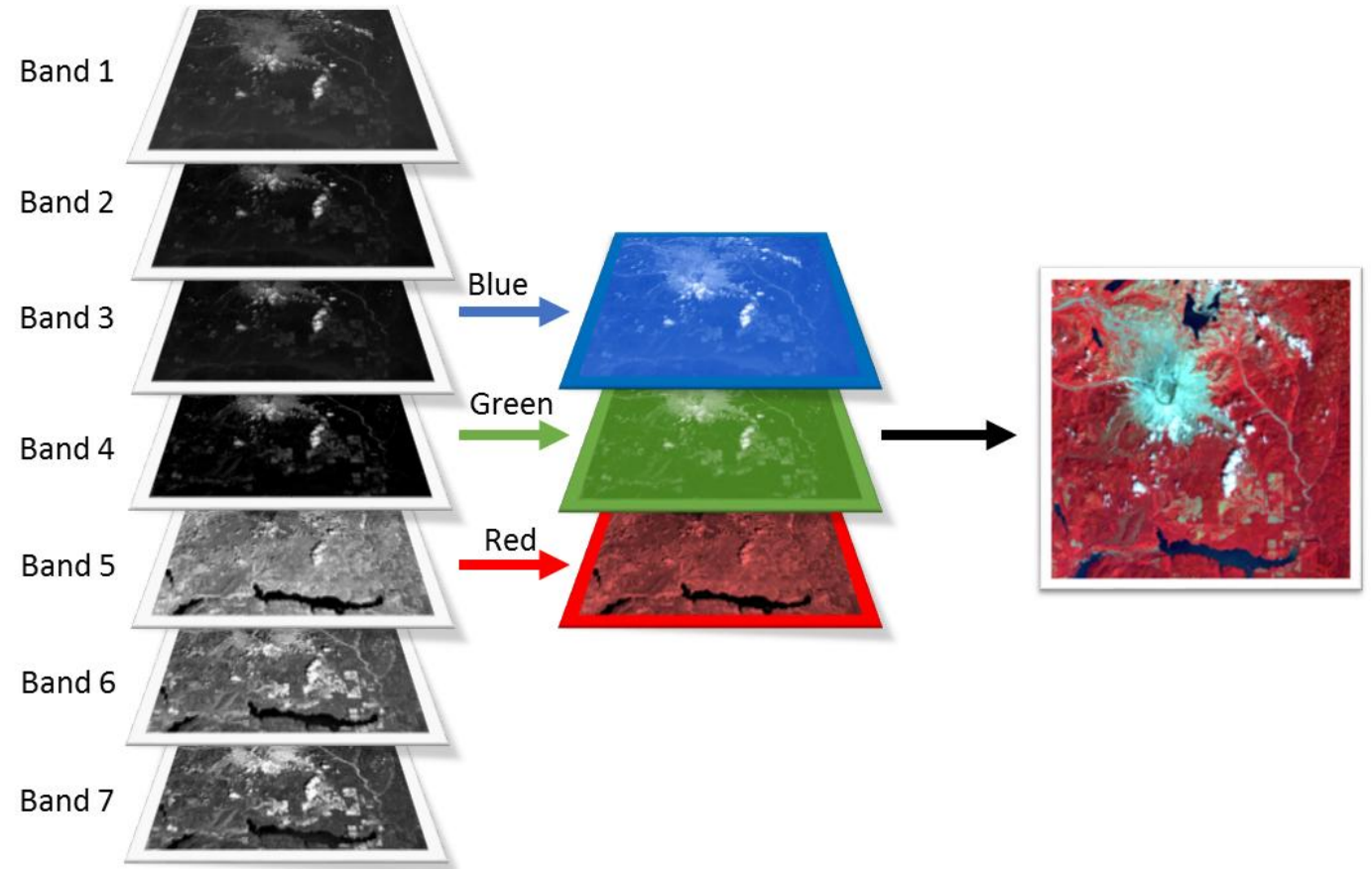


Landsat 8 Spectral Bands

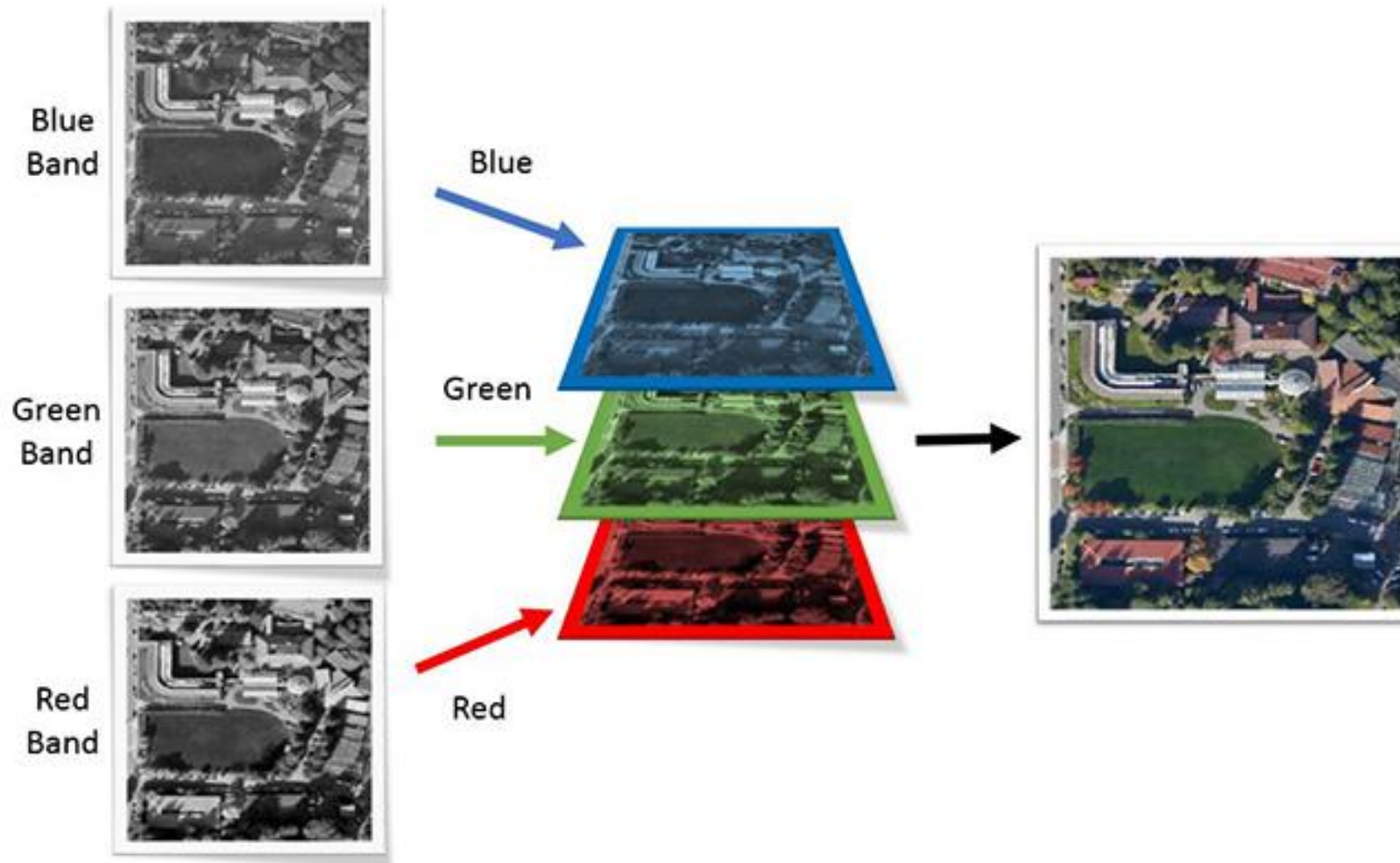
Landsat 8 Operational Land Image (OLI) and Thermal Infrared Sensor (TIRS)		
Band	Wavelength	Useful for mapping
Band 1 - coastal aerosol	0.43-0.45	Riset daerah pesisir dan aerosol
Band 2 - blue	0.45-0.51	Pemetaan batimetri (bathymetric), membedakan tanah dari tumbuh-tumbuhan dan tumbuhan gugur dari tumbuhan jenis konifera
Band 3 - green	0.53-0.59	Menentukan vegetasi puncak, yang berguna untuk menilai kekuatan tanaman
Band 4 - red	0.64-0.67	Membedakan lereng vegetasi
Band 5 - Near Infrared (NIR)	0.85-0.88	Menentukan kandungan biomassa dan garis pantai
Band 6 - Short-wave Infrared (SWIR) 1	1.57-1.65	Membedakan kadar air tanah dan tumbuh-tumbuhan; menembus awan tipis
Band 7 - Short-wave Infrared (SWIR) 2	2.11-2.29	Peningkatan kadar air tanah dan tumbuh-tumbuhan; menembus awan tipis
Band 8 - Panchromatic	0.50-0.68	Resolusi 15 meter, definisi gambar lebih tajam
Band 9 - Cirrus	1.36-1.38	Peningkatan deteksi kontaminasi awan cirrus
Band 10 - TIRS 1	10.60-11.19	Resolusi 100 meter, pemetaan termal dan perkiraan kelembaban tanah
Band 11 - TIRS 2	11.50-12.51	Resolusi 100 meter, pemetaan termal yang lebih baik, dan perkiraan kelembaban tanah

Band Combinations

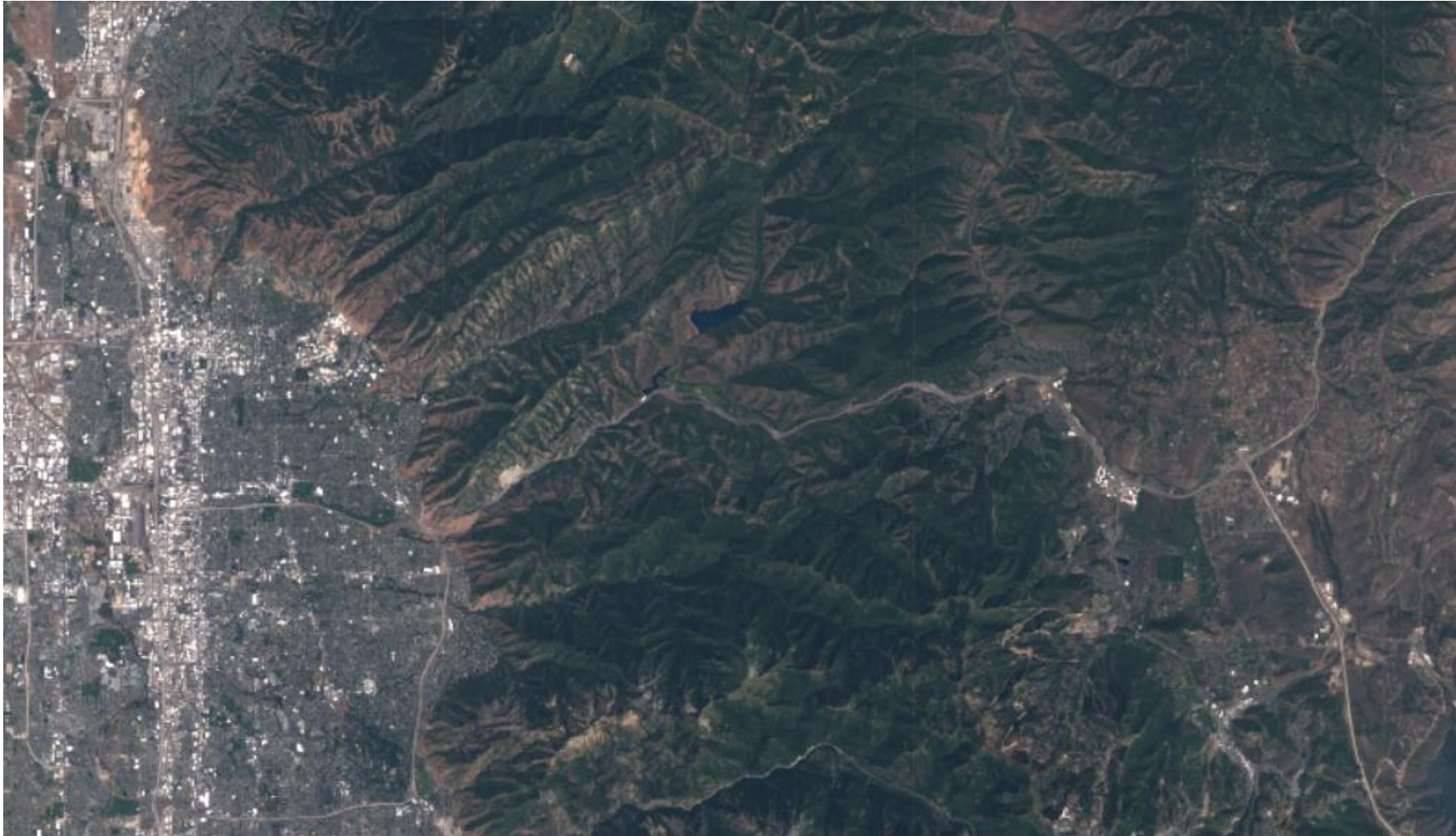
- Ketika menampilkan citra remote sensing dalam komputer, kita dibatasi hanya dapat menampilkan 3 band
- Inilah yang menyebabkan hasil komposit warna asli dan komposit warna palsu



Band Combinations :: Komposit Warna Asli



Landsat 8 :: Komposit warna asli



Band Combinations :: Komposit Warna Palsu

- Citra ditampilkan menggunakan semua band ketimbang Warna asli (RGB)
- Memungkinkan pengguna untuk memvisualisasikan panjang gelombang yang tidak dapat dilihat oleh mata manusia
- Menerapkan banyak kombinasi berbeda, bergantung pada aplikasinya

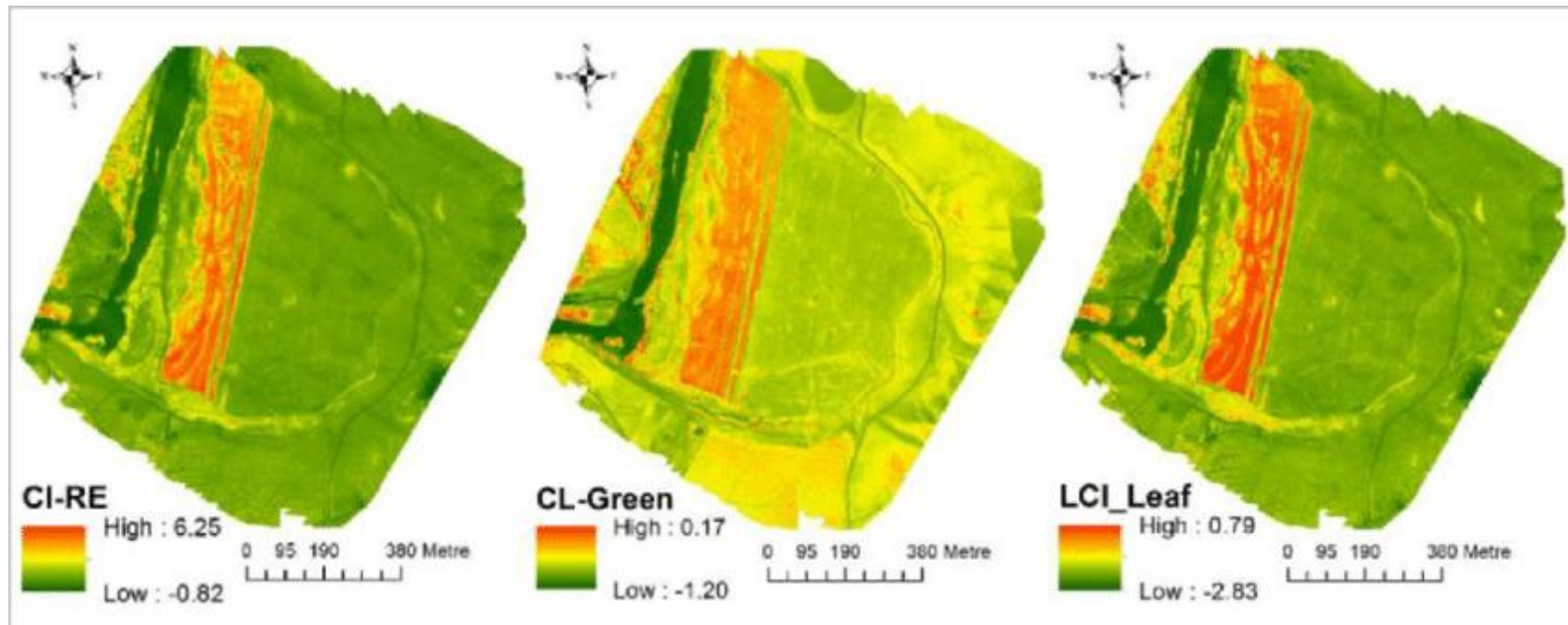


- Kombinasi Band Landsat 8 yang sering digunakan

Composite Name	Bands
Natural Color	4 3 2
False Color (urban)	7 6 4
Color Infrared (vegetation)	5 4 3
Agriculture	6 5 2
Healthy Vegetation	5 6 2
Land/Water	5 6 4
Natural With Atmospheric Removal	7 5 3
Shortwave Infrared	7 5 4
Vegetation Analysis	6 5 4

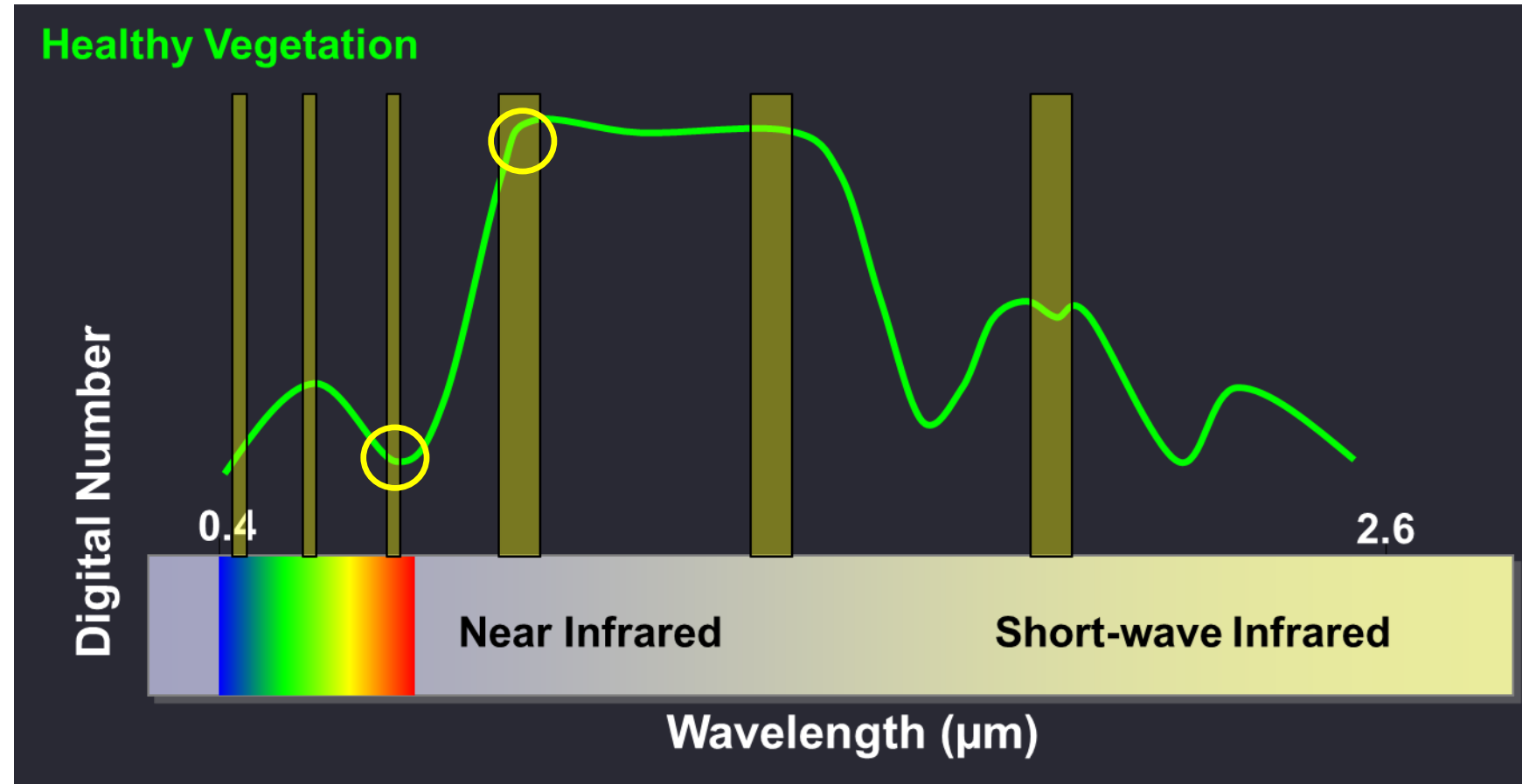
Rasio Band

- Indeks spektral sederhana yang dihitung dengan membagi satu band dengan band lainnya (matematika band)
- Digunakan untuk meningkatkan perbedaan spektral antar band
- Sangat berguna ketika mengidentifikasi jenis lahan



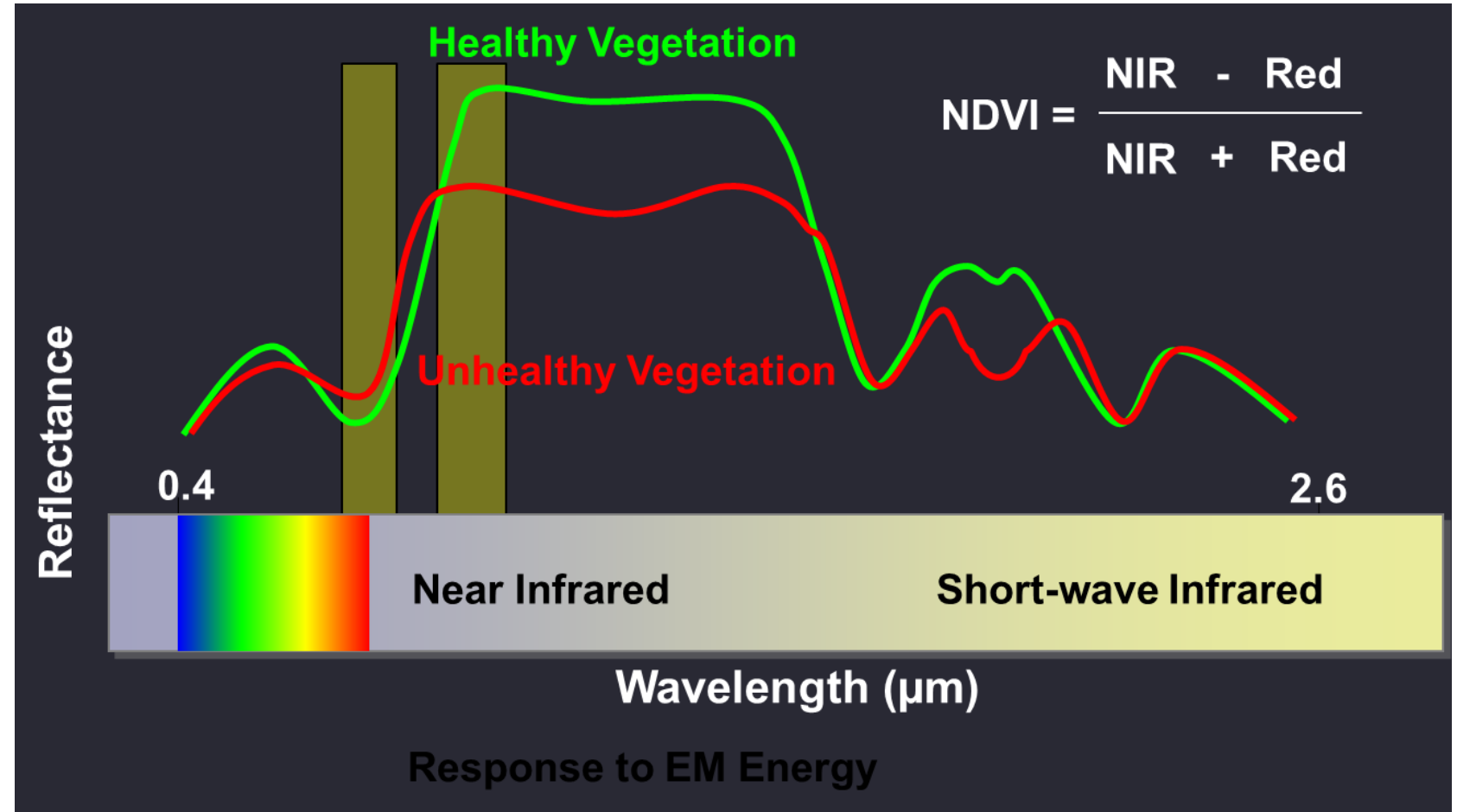
Rasio Band

- Ciri spektral vegetasi sehat memiliki puncak dan lembah yang jelas yang ditangkap dalam pita sensor



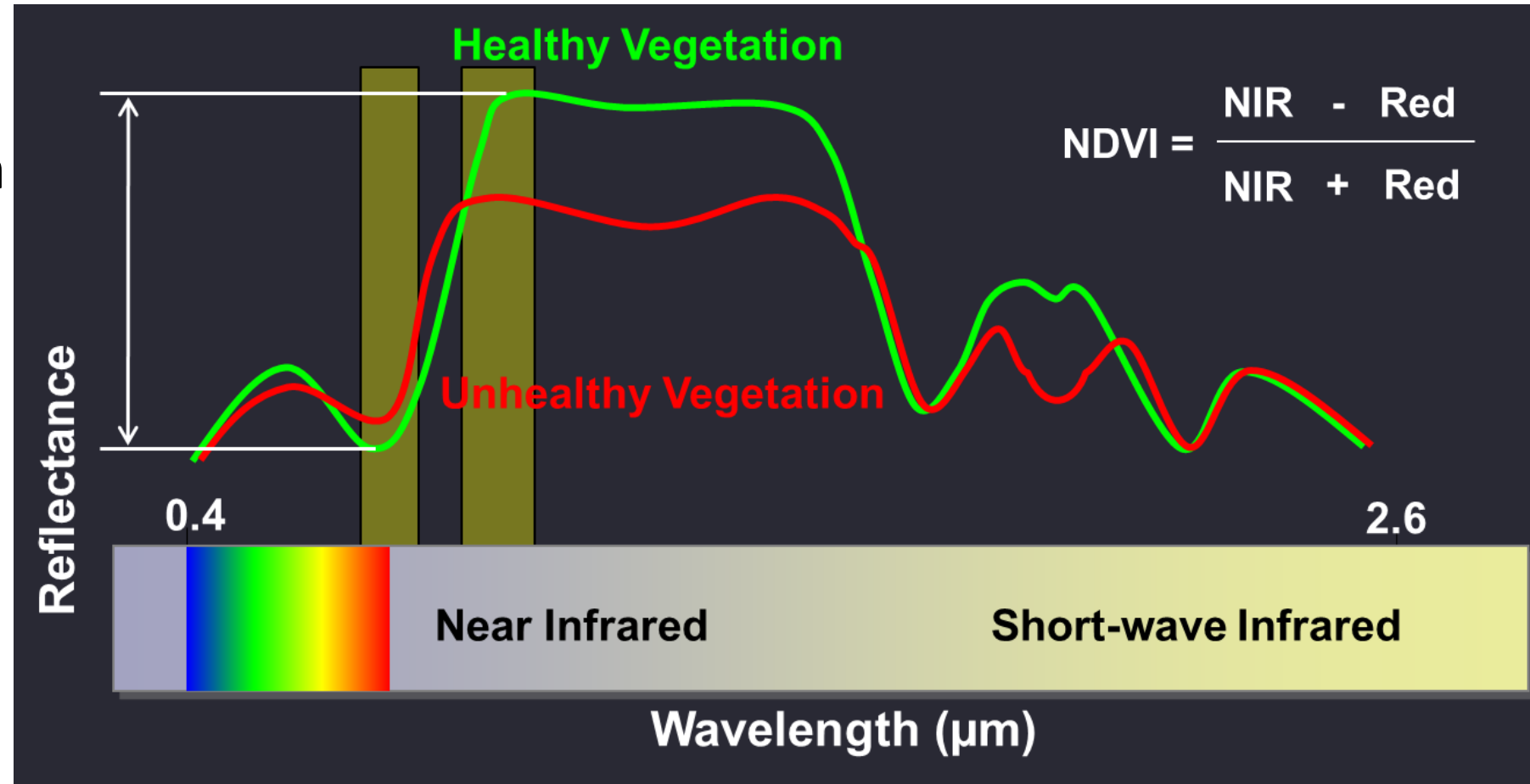
Rasio Band : NDVI

- NDVI: Normalized Difference Vegetation Index
- Indeks ini memungkinkan kita membedakan antara vegetasi yang sehat dan tidak sehat



Rasio Band : NDVI

- NDVI untuk vegetasi sehat berbeda dengan vegetasi tidak sehat



Index Vegetasi : NDVI

- Metode standar untuk membandingkan Tingkat kehijauan tanaman (vegetasi) dari satelit
- Index NDVI bernilai: -1 s.d +1
- Index mendekati +1 dinyatakan sebagai vegetasi **kondisi sehat**

$$\text{NDVI} = \frac{\text{NIR} - \text{RED}}{\text{NIR} + \text{RED}}$$

HEALTHY
VEGETATION REFLECTANCE

50% NIR 8% RED



NDVI = 0.72

STRESSED
VEGETATION REFLECTANCE

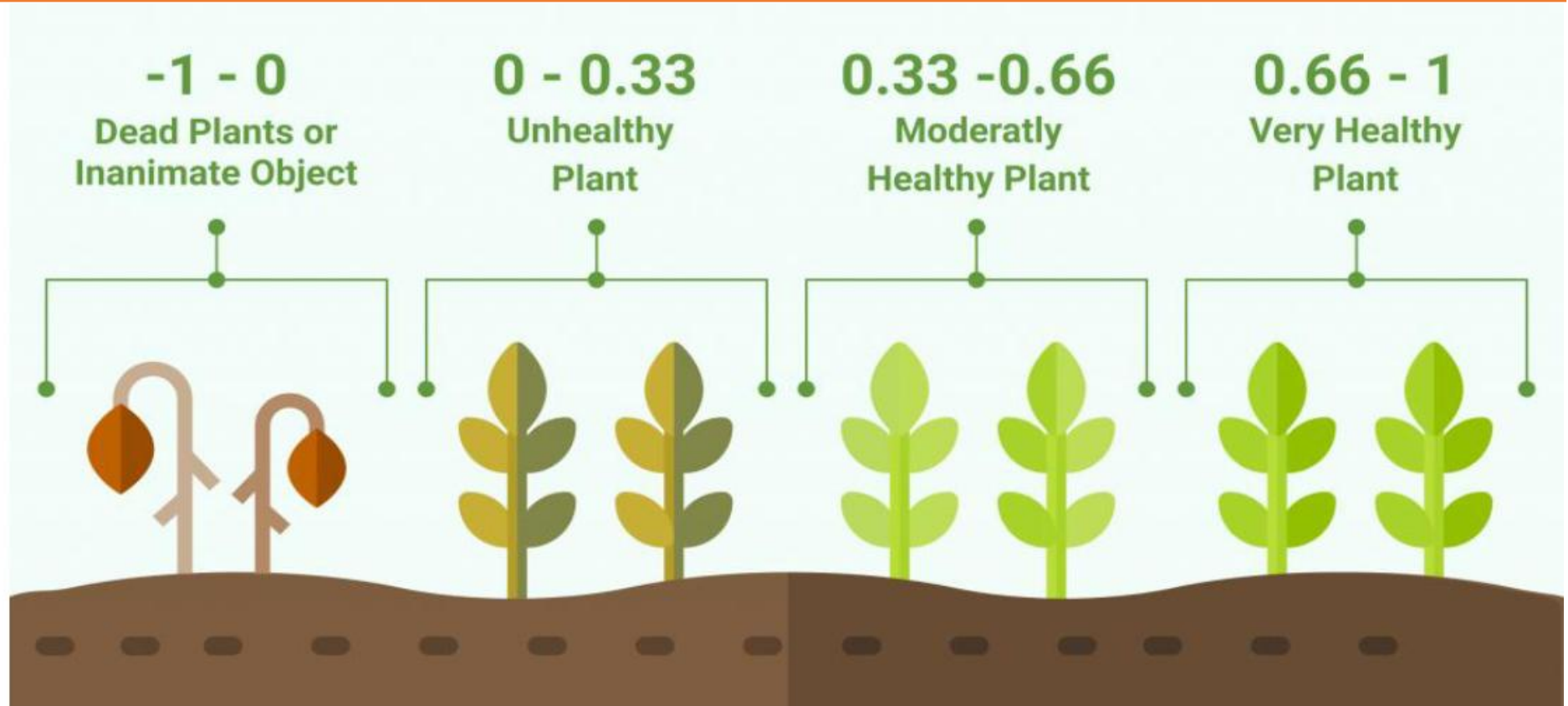
40% NIR 30% RED



NDVI = 0.14

Index Vegetasi : NDVI

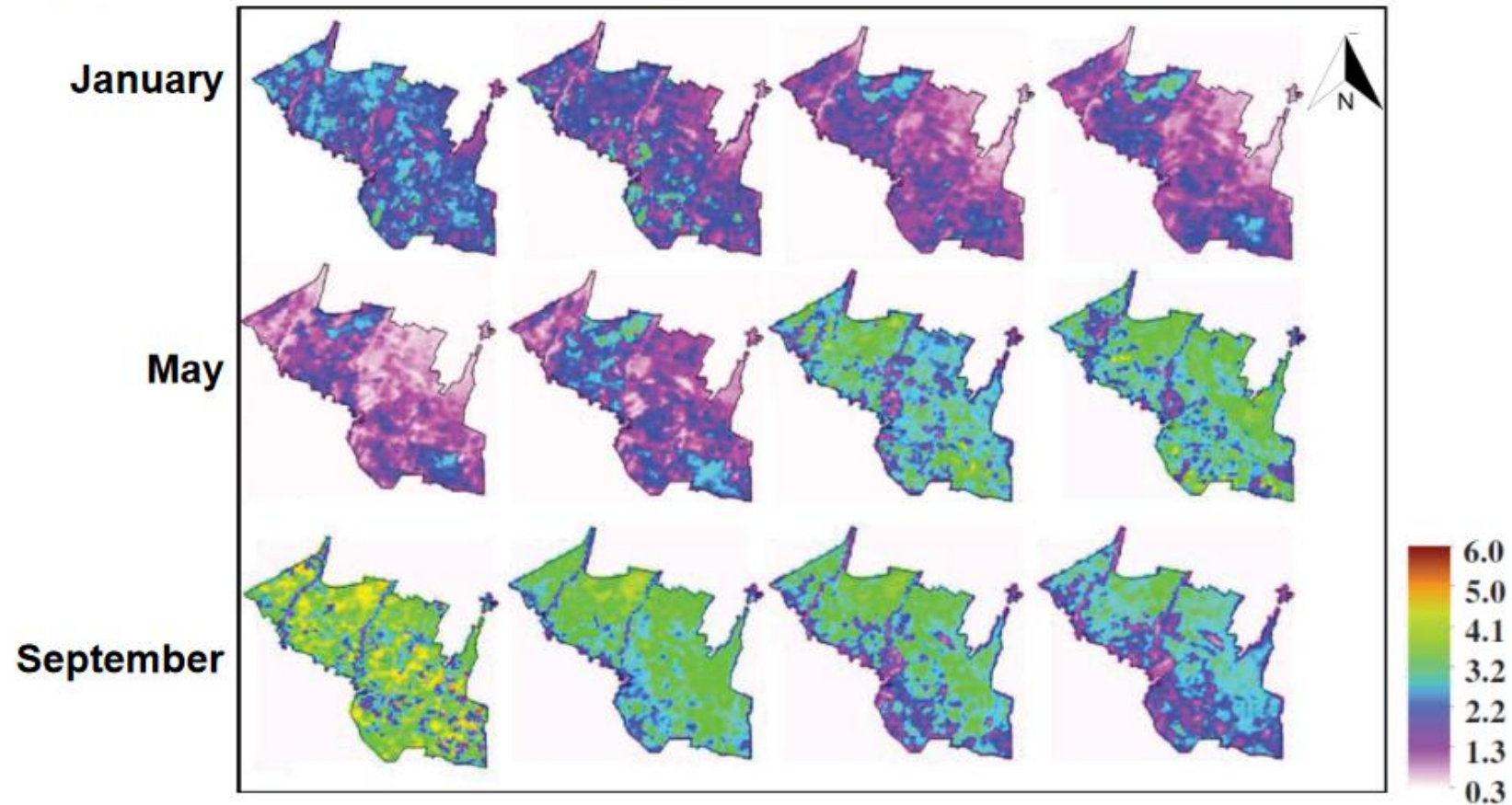
NDVI



- Nilai negatif beasosiasi: **permukaan air, struktur buat manusia, batu, awan, salju**
- **Tanaman** selalu bernilai positif antara **0.2 s.d 1**

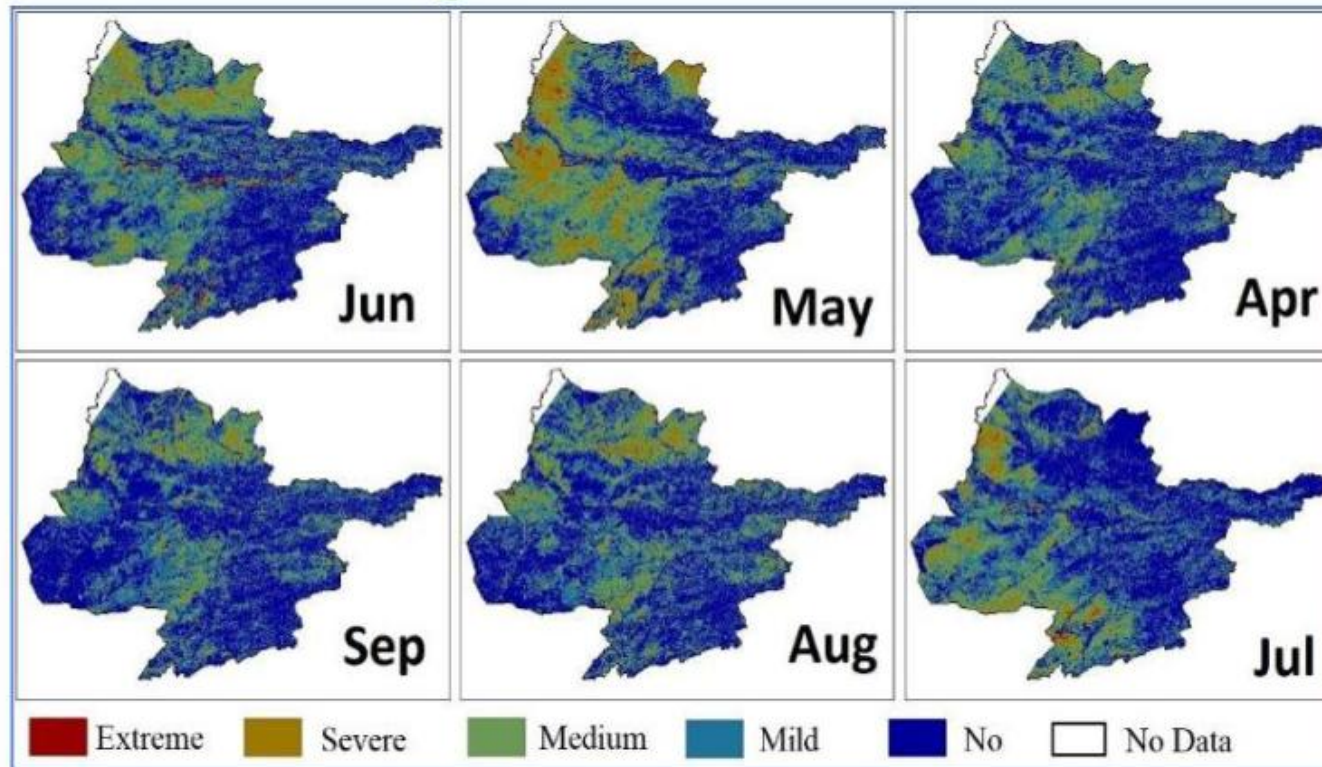
Index Remote Sensing: Vegetasi

Index Name	Equation	Purpose/ Application
NDVI (Normalized Difference Vegetation Index)	$((\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red}))$	Standardized index to generate an image displaying greenness (relative biomass)
SAVI (Soil-Adjusted Vegetation Index)	$((\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red} + L)) \times (1 + L)$	Vegetation index that attempts to minimize soil brightness influences. L = amount of green veg. cover
VARI (Visible Atmospherically Resistant Index)	$(\text{Green} - \text{Red}) / (\text{Green} + \text{Red} - \text{Blue})$	Emphasized vegetation in the visible portion of the spectrum, while mitigating illumination differences and atmospheric effects



Monthly pattern (January- December) of leaf area index (LAI) derived from NDVI (exponential relationship)

Aplikasi Index Vegetasi



Monthly median of drought condition in Herat province during vegetation seasons of 2003-2014 based on VCI

Source: Mohammad Ehsan Razipoor, 2019

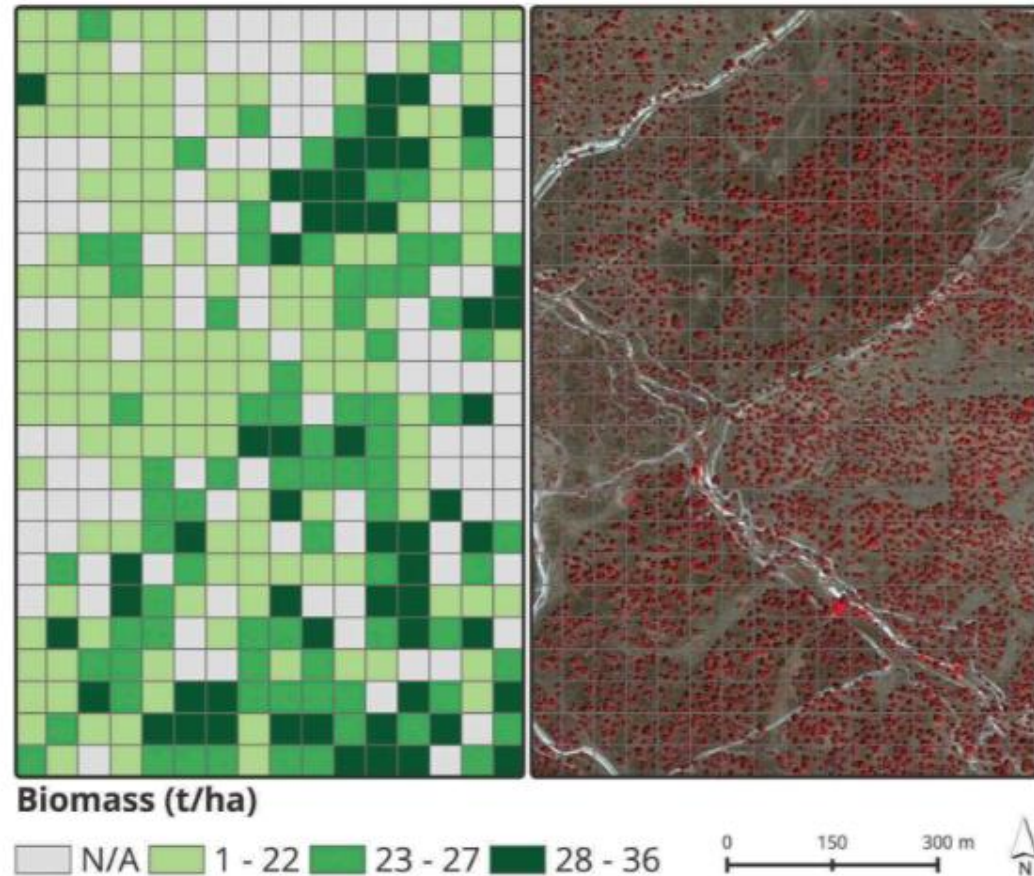
http://www.fao.org/giews/earthobservation/asis/index_2.jsp?lang=en

Biomass estimation

	AGB (t/ha)	CC (%)
AGB (t/ha)	1	
CC (%)	0.83	1
NDVI	0.85	0.95
EVI	0.75	0.91
SR	0.86	0.96
SAVI	0.70	0.84

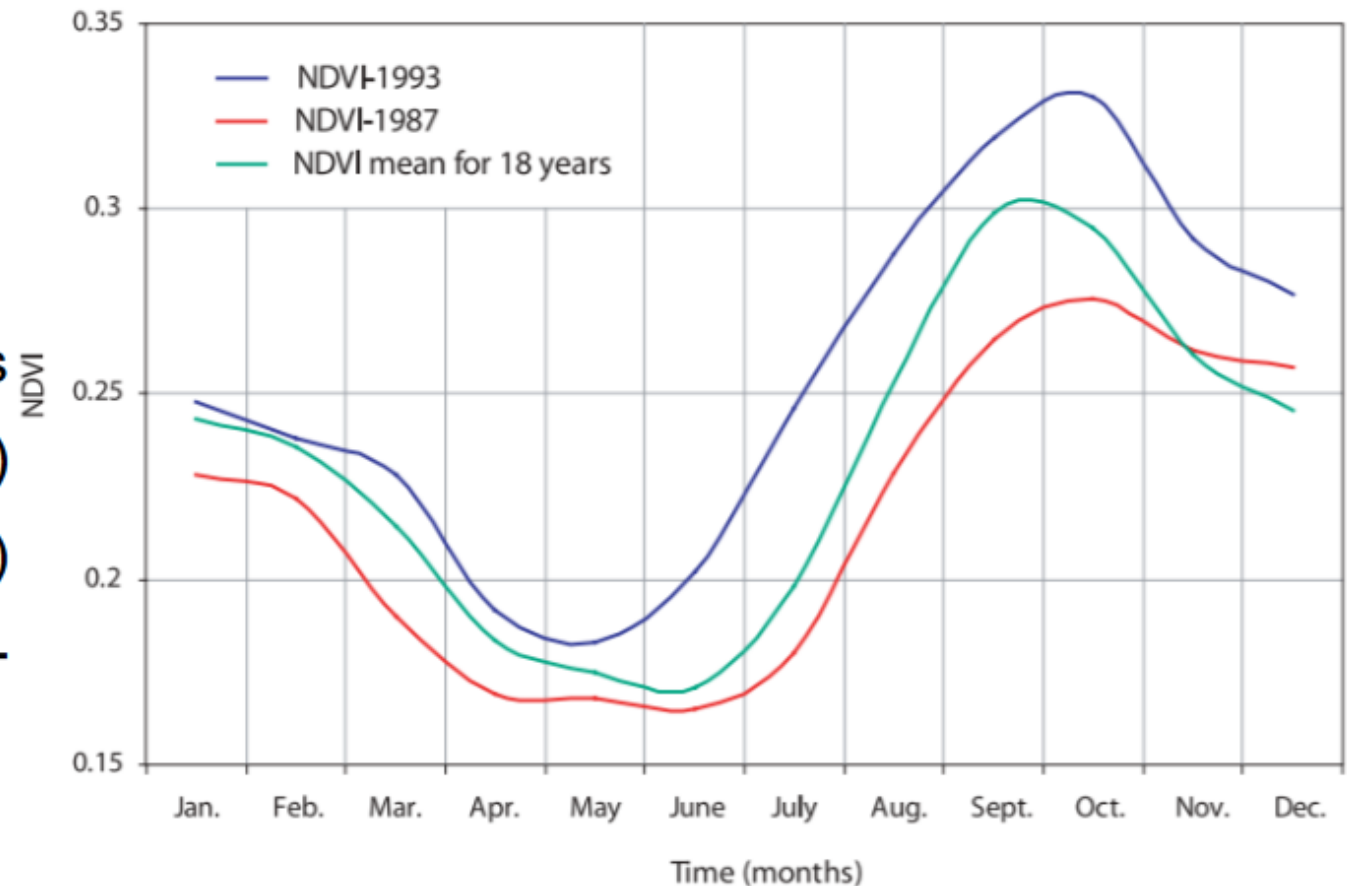
Correlation between above ground biomass and vegetation indices for *Quercus rotundifolia*

Source: Macedo et al., 2018, southern Portugal



Southwest Asia NDVI deviation

- A monthly NDVI time series for a drought year (1987) and a wet year (1993) compared to the NDVI long-term mean

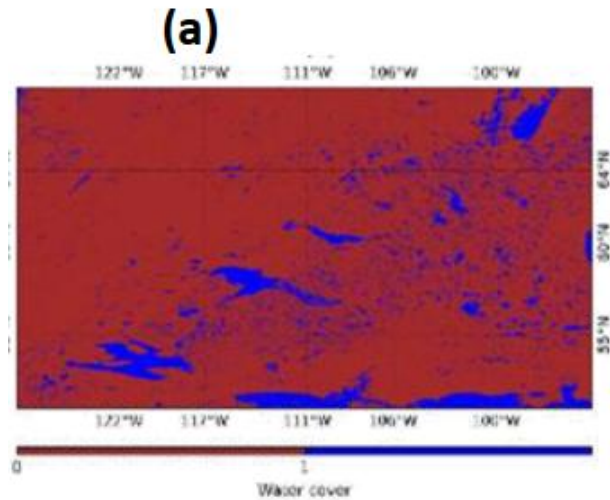


Source: Thenkabail and Gamage, 2004

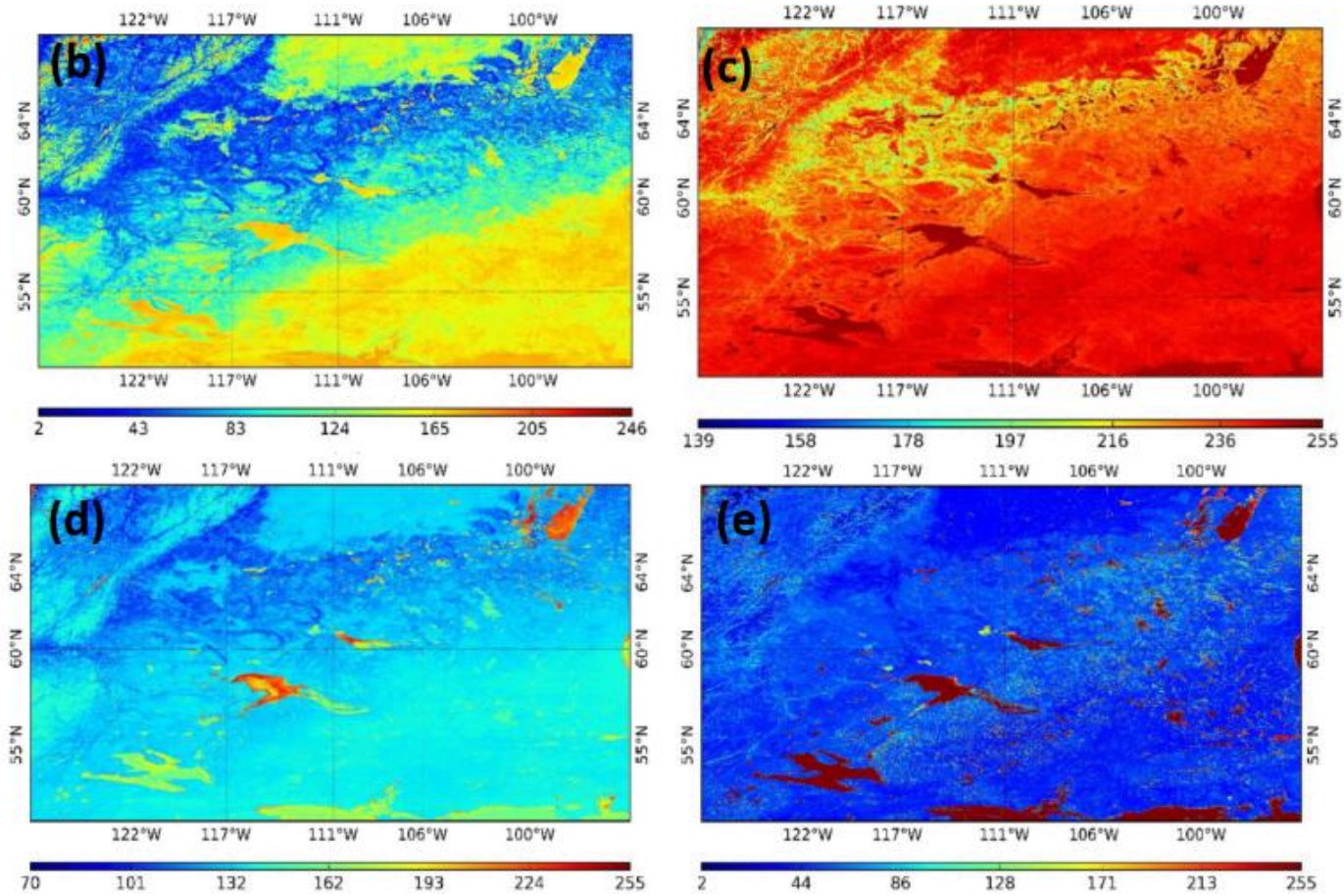
Index Remote Sensing: Air

Index Name	Equation	Purpose/ Application
NDSI (Normalized Difference Snow Index)	$(\text{Green} - \text{SWIR}) / (\text{Green} + \text{SWIR})$	Identifies snow cover while ignoring cloud cover. Designed for MODIS and Landsat data.
MNDWI (Modified Normalized Difference Water index)	$(\text{Green} - \text{SWIR}) / (\text{Green} + \text{SWIR})$	Enhances open water features.
NDMI (Normalized Difference Moisture Index)	$(\text{NIR} - \text{SWIR1}) / (\text{NIR} + \text{SWIR1})$	Sensitive to moisture levels in vegetation. Used to monitor droughts and fuel levels in fire prone areas.

Water mapping

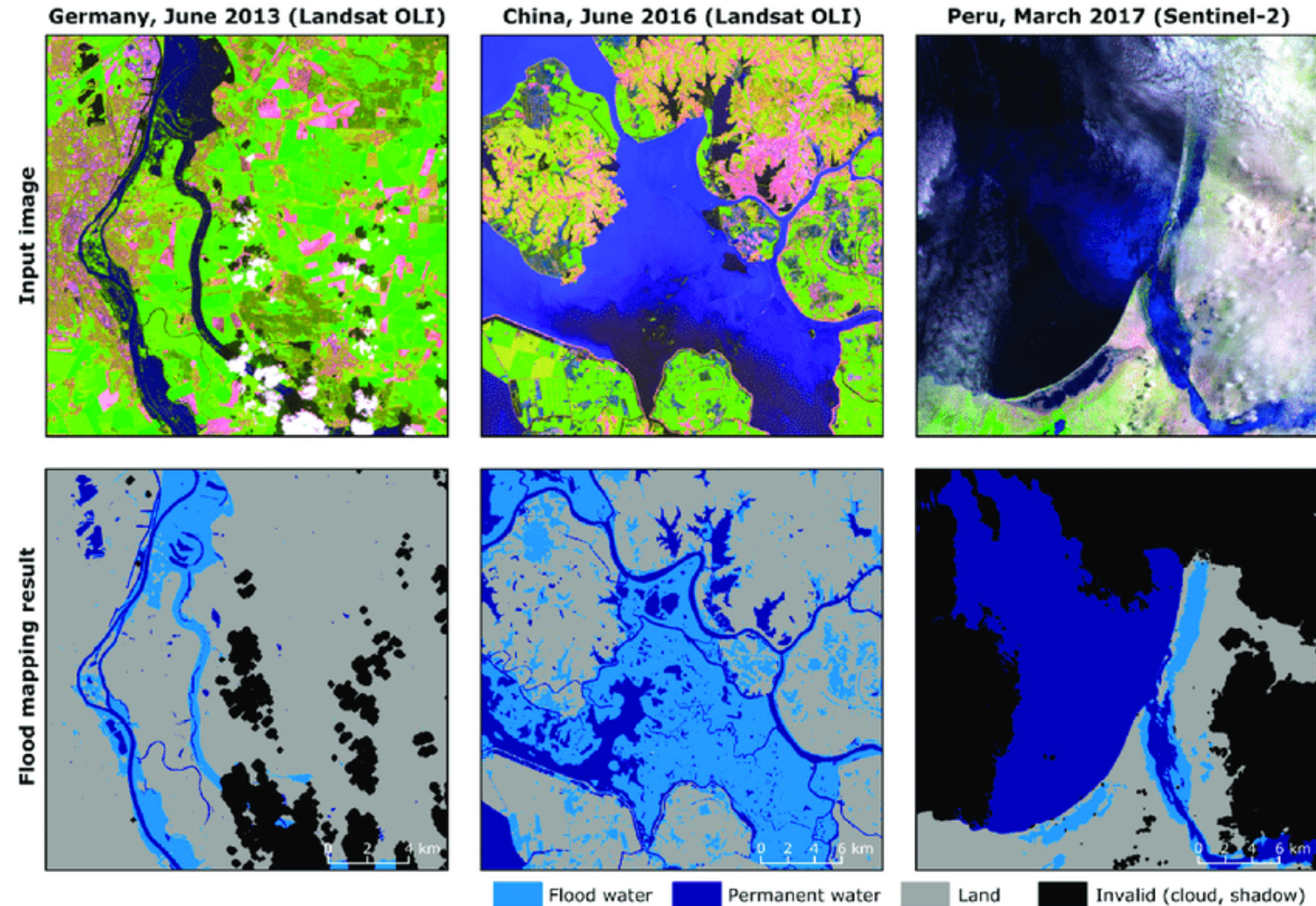


Source: Sharma et al, 2015



(a) Reference water cover map (b) Automated water extraction index (AWEI), (c) Modified normalized difference water index (MNDWI) (d) Normalized difference water index (NDWI), (e) Superfine water index (SWI)

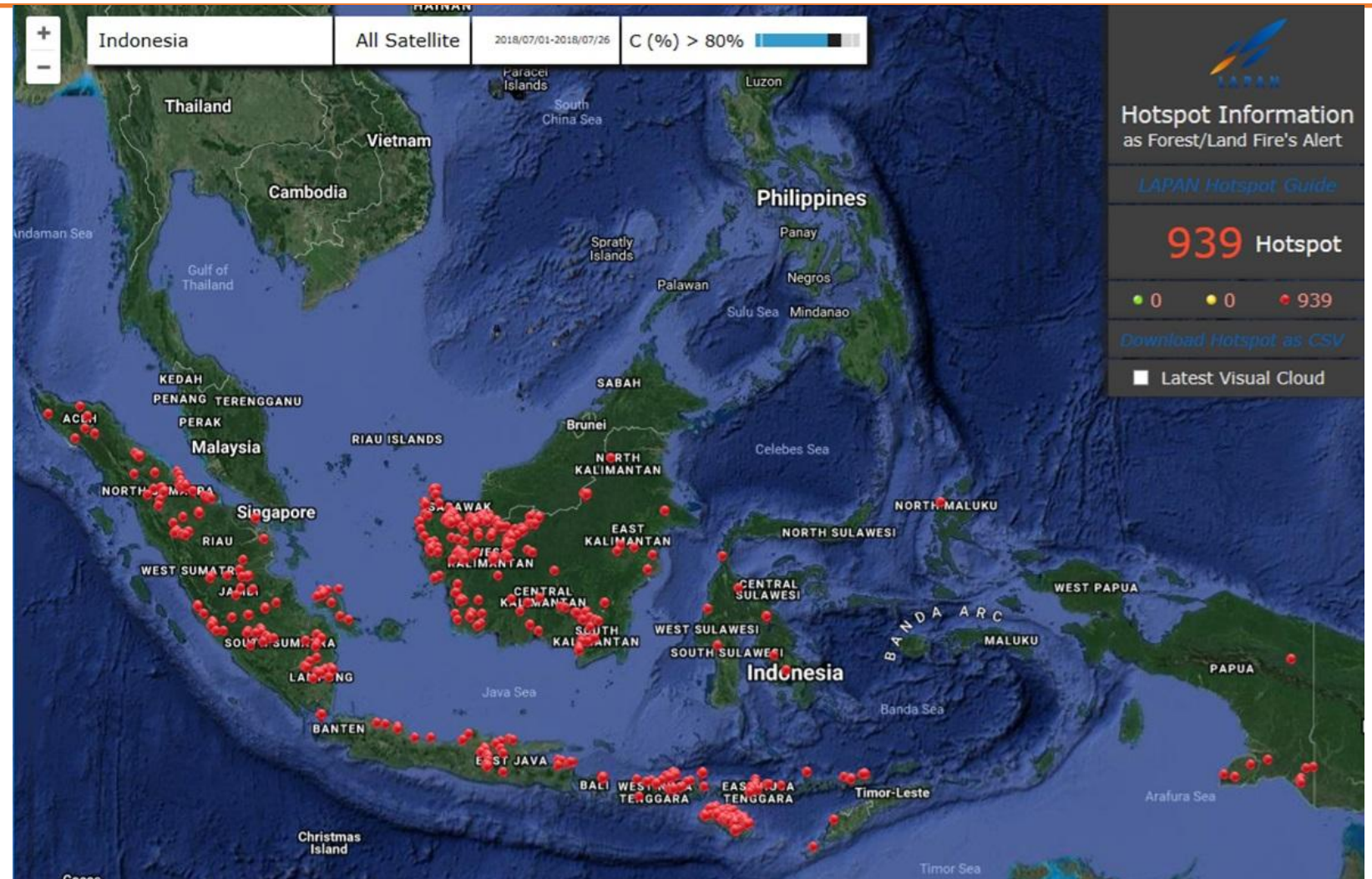
Flood Mapping



Index Remote Sensing: Landscape

Index Name	Equation	Purpose/ Application
BAI (Burn Area Index)	$1/((0.1 - \text{RED})^2 + (0.06 - \text{NIR})^2)$	Identify areas of terrain affected by fire.
NBR (Normalized Burn Ratio Index)	$(\text{NIR} - \text{SWIR}) / (\text{NIR} + \text{SWIR})$	Emphasized burned areas, while mitigating illumination and atmospheric effects.
NDBI (Normalized Difference Built-up Index)	$(\text{SWIR} - \text{NIR}) / (\text{SWIR} + \text{NIR})$	Emphasized man-made built-up areas. Mitigates effects of terrain illumination and atmospheric effects.

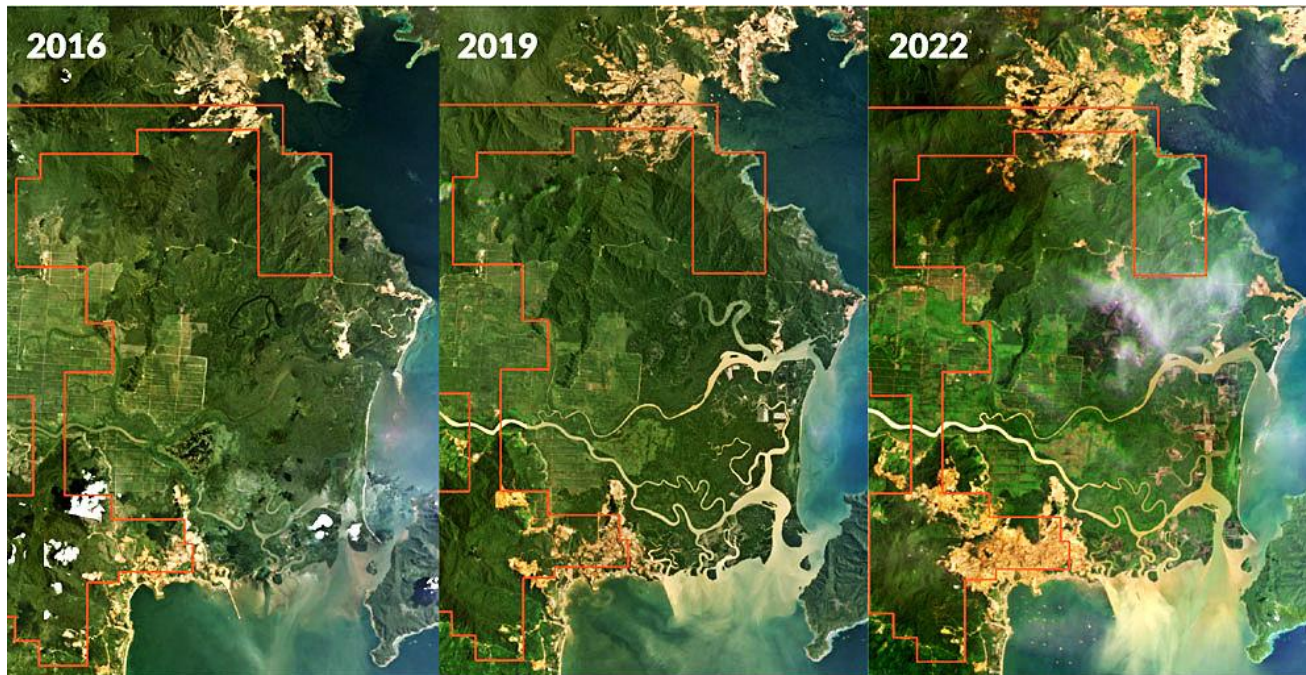
Peta sebaran asap Kebakaran



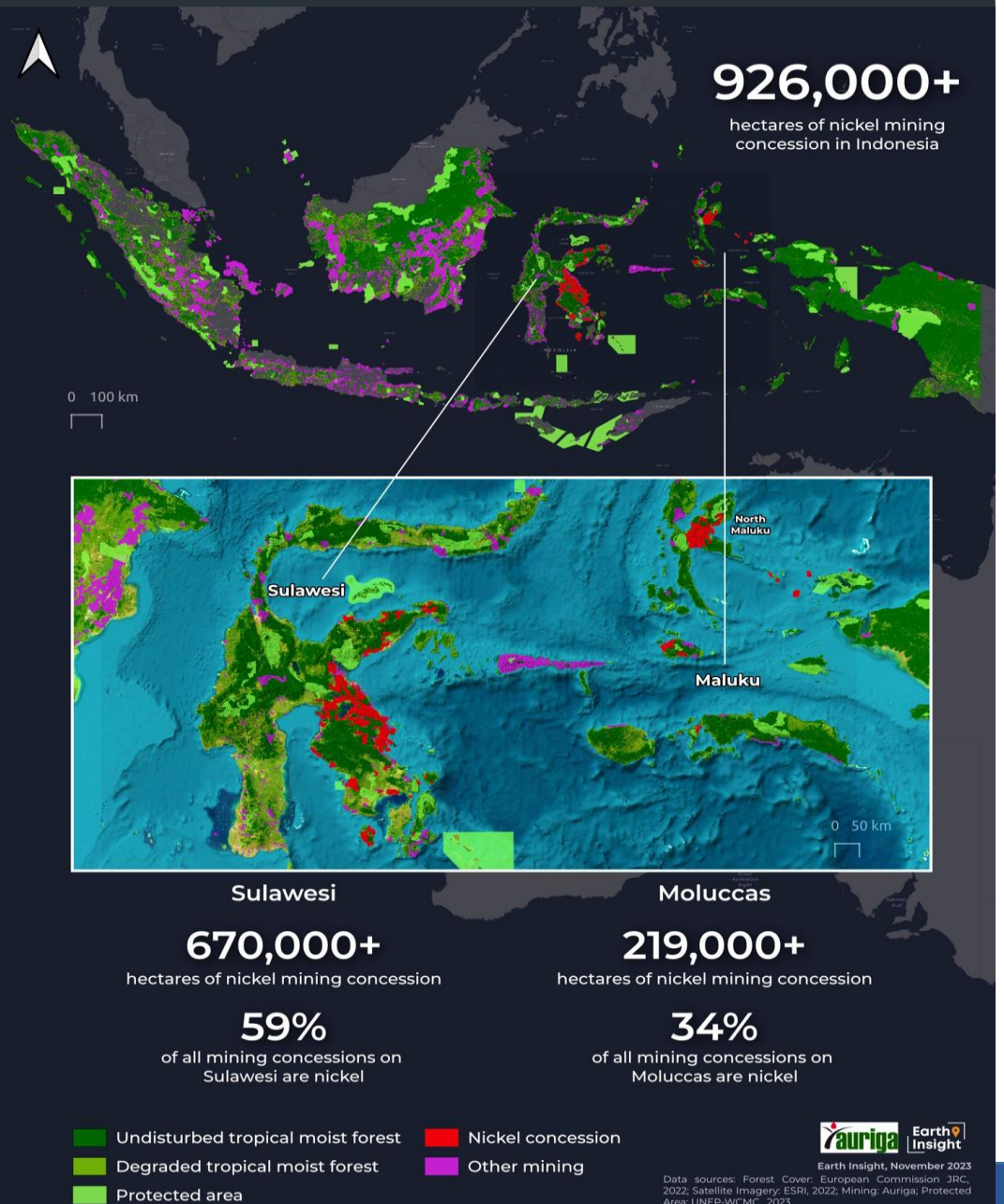
Index Remote Sensing: Geology

Index Name	Equation	Purpose/ Application
Clay Minerals Ratio	$\text{SWIR1} / \text{SWIR2}$	Leverages the fact that hydrous minerals absorb radiation in 2.0-2.3 micron portion of EM spectrum.
Ferrous Minerals Ratio	SWIR / NIR	Highlights iron-bearing minerals.
Carbonate Index	$(\text{red} - \text{green}) / (\text{red} + \text{green})$	
Rock Outcrop Index	$(\text{SWIR1} - \text{green}) / (\text{SWIR1} + \text{green})$	

Aplikasi Index Geology



Indonesia: Nickel Mining





Praktikum

- Menggunakan Aplikasi QGIS
- Explore GEE untuk pengolahan citra satelit